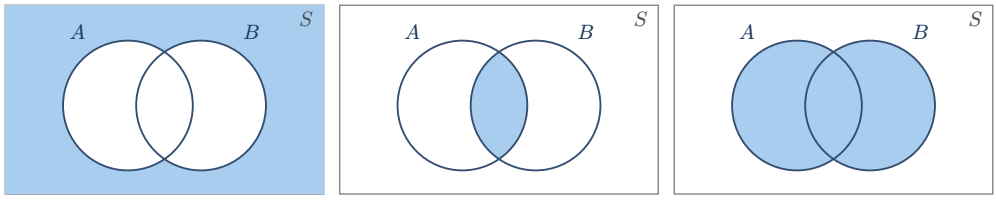
PROBABILITY AND STATISTICS I

Probability Axioms

Lecture 03

Set operations · Venn diagrams · the axioms and their theorems



Lecture notes with every worked example solved,
vector Venn diagrams, short proofs of every theorem,
and practice problems for independent work.

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Previous chapter: Sample Spaces and Probability

Contents

1 Set operations 3
1.1 Sets as events 4
2 Venn diagrams 5
3 Identities 5
4 Axioms of probability 6
5 Properties of probability 7
5.1 Worked example 8
6 Mutually exclusive events 9
7 Working with the rules 10
7.1 The method, in three lines 12

1 Set operations

Chapter 2 made an **event** a subset of the sample space. That one decision means every tool for combining sets is also a tool for combining events — "A or B", "both", "not A" — and this lecture builds probability on top of exactly those tools. Throughout, S is the sample space and every set is a subset of it.

Definition 3.1 — Equality. Two sets A and B are **equal**, denoted $A = B$, if and only if $A \subset B$ and $B \subset A$.

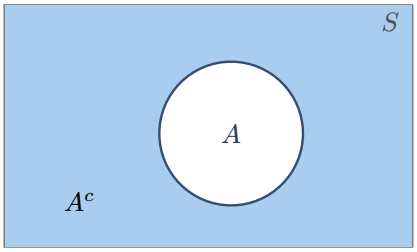
Note. In this course $A \subset B$ means "every element of A is in B", and allows $A = B$ — what some books write $A \subseteq B$. With the strict reading, the definition above would be impossible to satisfy. In practice: to show two sets are equal, show each is inside the other.

Definition 3.2 — Complement. Suppose $A \subset S$. The **complement** of A, denoted A^c (or $\overline{A}$), is the set containing all elements in S but not in A:

$$A^c = \{\zeta : \zeta \in S \text{ and } \zeta \notin A\}.$$

Example 1. If the sample space is $S = \{1, 2, 3, 4, 5\}$ and $A = \{1, 2, 4\}$ is an event on S, find A^c.
Solution. Strike the elements of A out of S and keep what is left:

$$A^c = \{3, 5\}.$$



Definition 3.3 — Union. The **union** of A and B, denoted $A \cup B$, is the set containing all elements in either A or B or both:

$$A \cup B = \{\zeta : \zeta \in A \text{ or } \zeta \in B\}.$$

Read it as **at least one**. The "or" is inclusive — an element in both sets is in the union.

Example 2. If $A = \{1, 2, 5\}$ and $B = \{2, 3, 6\}$, find $A \cup B$.
Solution. Collect everything that appears in either list, writing the shared element 2 only once:

$$A \cup B = \{1, 2, 3, 5, 6\}.$$

The order of the elements in a set does not matter; listing them in increasing order just makes it easier to check that nothing is missing or repeated.

Definition 3.4 — Intersection. The **intersection** of A and B, denoted $A \cap B$, is the set containing all elements in both A and B:

$$A \cap B = \{\zeta : \zeta \in A \text{ and } \zeta \in B\}.$$

Read it as **both**.

Example 3. If $A = \{1, 2, 5\}$ and $B = \{2, 3, 6\}$, find $A \cap B$.
Solution. Only 2 is in both lists:

$$A \cap B = \{2\}.$$

Definition 3.5 — Null set. The set containing no element is called the **null set**, denoted Φ. Note that

$$\Phi = S^c.$$

Definition 3.6 — Difference. The **difference** $B - A$ is the set of elements in B but not in A:

$$B - A = B \cap A^c.$$

Note. Definition 3.6 is not in the original handout, but Theorem 6 and Corollary 1 use $B - A$ and $B \cap A^c$, and they are the same set. Writing it once here saves a surprise later.

1.1 Sets as events

Every set operation has a meaning in the language of events. This table is the dictionary you translate word problems through — most of the errors in this chapter are translation errors, not arithmetic ones.

Set	Event — in words
S	the **certain** event — it always occurs
Φ	the **impossible** event — it never occurs
A^c	A did **not** occur
$A \cup B$	**either** A or B **or both** occurred — at least one
$A \cap B$	**both** A and B occurred
$A \cap B^c = A - B$	A occurred **but not** B — "only A"
$(A \cup B)^c = A^c \cap B^c$	**neither** A nor B occurred
$(A \cap B^c) \cup (A^c \cap B)$	**exactly one** of A, B occurred

The first five lines are the original lecture's remark; the last three are combinations of them that appear in almost every exam, and are worth recognising on sight.

Exercise set 1

1.1 Let $S = \{1, 2, ..., 10\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{2, 4, 6, 8, 10\}$. Find A^c, $A \cup B$, $A \cap B$, $A \cap C$, $(A \cup B)^c$ and $B - A$.

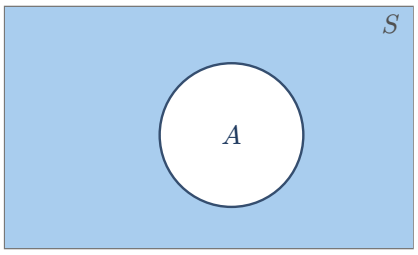
1.2 A die is tossed once. Let E be "an even number" and F be "a number greater than 3". List $E \cap F$, $E \cup F$ and $E^c \cap F$, and say each in words.

1.3 Write in set notation: (a) neither A nor B occurs; (b) A occurs but B does not; (c) exactly one of A, B occurs; (d) at least one of A, B, C occurs.

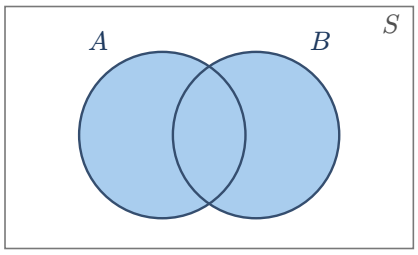
1.4 With the sets of question 1, check that $(A \cap B)^c$ and $A^c \cup B^c$ are the same set by listing both.

2 Venn diagrams

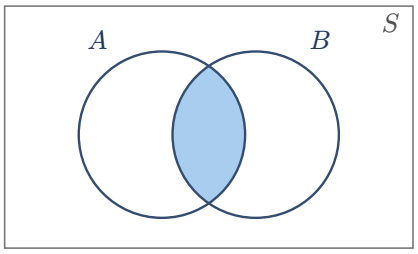
A graphical representation that is very useful for illustrating set operations is the **Venn diagram**. The rectangle is the sample space S; each event is a region inside it; the shaded part is the set being described.



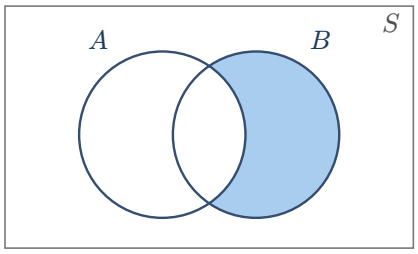
A^c — everything outside A



$A \cup B$ — at least one



$A \cap B$ — both



$B \cap A^c = B - A$ — only B

Note. Every two-event Venn diagram has exactly **four** regions — only A, both, only B, neither — and every set built from A and B is a union of some of them. Naming the set means naming its regions.

Watch out. A Venn diagram is a picture of **which outcomes** are in a set, not of **how likely** they are. The size of a region on the page means nothing: a large circle can have probability 0.01. Use the diagram to decide which regions you need, then use the theorems to compute them.

3 Identities

By the definitions above we obtain the following identities. Each one can be read straight off a Venn diagram.

Identity	Why
$S^c = \Phi$	Nothing in S is outside S.
$\Phi^c = S$	Everything in S is outside the empty set.
$S \cup A = S$	A adds nothing that S does not already contain.
$S \cap A = A$	Every element of A is already in S.
$S \cup A^c = S$	The same line again, with A^c in place of A.
$S \cap A^c = A^c$	Likewise.

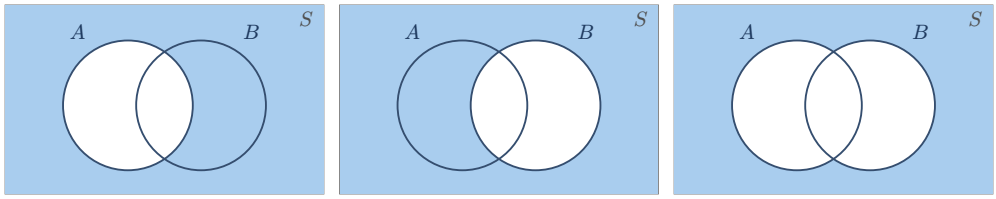
Rule 3.1 — Commutative laws.

$$A \cup B = B \cup A, \qquad A \cap B = B \cap A.$$

Rule 3.2 — De Morgan's laws.

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

De Morgan's laws are the ones that matter in practice. In words: **not (at least one)** is **neither**, and **not (both)** is **at least one fails**. The first law in pictures: shade A^c, shade B^c, and keep only what is shaded in both — what survives is exactly the outside of $A \cup B$.



A^c B^c $A^c \cap B^c = (A \cup B)^c$

Note. Two more laws are used without comment in the proofs below, and hold for exactly the same reasons: **associative** — $(A \cup B) \cup C = A \cup (B \cup C)$, and the same for $\cap$; **distributive** — $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ and $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

4 Axioms of probability

Chapter 2 defined probability by counting: $P(A) = n/N$, valid only when the outcomes are equally likely. The axiomatic definition drops that condition. It does not say **how** to find $P(A)$ — it says what any assignment of probabilities must obey to deserve the name.

Let S be a finite sample space and A an event in S. In the axiomatic definition, the probability $P(A)$ of the event A is a real number assigned to A which satisfies the following three axioms:

Axioms of probability.

Axiom 1.	$P(A) \geq 0$	probabilities are never negative
Axiom 2.	$P(S) = 1$	something certainly happens
Axiom 3.	$P(A \cup B) = P(A) + P(B)$ whenever $A \cap B = \Phi$	disjoint events add

Note. The classical rule of Chapter 2 obeys all three: $n/N \geq 0$; $P(S) = N/N = 1$; and if A and B share no outcomes, their counts add. So $P = n/N$ is one **model** that satisfies the axioms — and everything proved from the axioms holds for it automatically.

Axiom 3 extends to any number of events by applying it repeatedly: if $A_1, A_2, ..., A_k$ are **pairwise** disjoint (no two share an outcome), then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k) = P(A_1) + P(A_2) + \cdots + P(A_k).$$

5 Properties of probability

By using the axioms, the following useful properties of probability can be obtained. Each proof is two or three lines; read them, because the **method** — split an event into disjoint pieces, then apply Axiom 3 — is the method for the exercises as well.

Theorem 1. For every event A,

$$0 \le P(A) \le 1,$$

i.e. a probability lies between 0 and 1.

Proof. $P(A) \ge 0$ is Axiom 1. For the upper bound, A and A^c are disjoint and $A \cup A^c = S$, so by Axioms 3 and 2, $P(A) + P(A^c) = 1$. Since $P(A^c) \ge 0$, $P(A) \le 1$. □

Theorem 2. For Φ, the empty set,

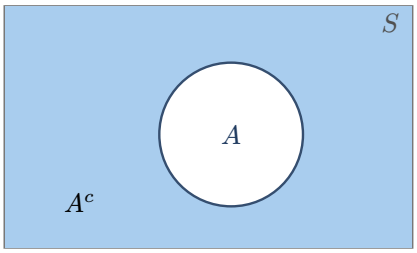
$$P(\Phi) = 0,$$

i.e. the impossible event has probability zero.

Proof. Φ is the complement of S, so by Theorem 3 below, $P(\Phi) = 1 - P(S) = 1 - 1 = 0$. (Or directly: $S = S \cup \Phi$ with $S \cap \Phi = \Phi$, so $P(S) = P(S) + P(\Phi)$.) □

Theorem 3. If A^c is the complement of A, then

$$P(A^c) = 1 - P(A).$$

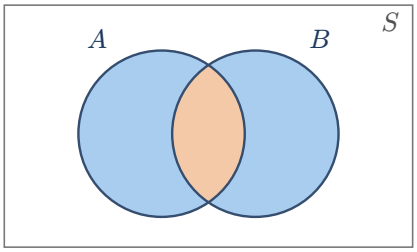


Proof. $A \cup A^c = S$ and $A \cap A^c = \Phi$. Axiom 3 then Axiom 2 give $P(A) + P(A^c) = P(S) = 1$; rearrange. □

Note. Theorem 3 is the **at least one** tool. "At least one head in five tosses" is 31 outcomes to count, or one to subtract: $1 - P(\text{no heads})$. When the event is hard and its complement is easy, compute the complement.

Theorem 4. (Addition rule.) If A and B are any two events, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$

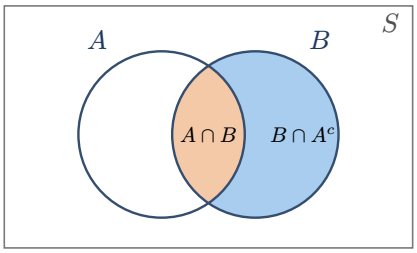


Proof. Split $A \cup B$ into the disjoint pieces A and $B \cap A^c$. By Axiom 3, $P(A \cup B) = P(A) + P(B \cap A^c)$, and by Corollary 1 below, $P(B \cap A^c) = P(B) - P(A \cap B)$. □

The subtraction is there because $P(A) + P(B)$ counts the overlap (amber) **twice** — once inside A, once inside B. Subtracting $P(A \cap B)$ once puts it back to being counted once.

Corollary 1. If A and B are any two events, then

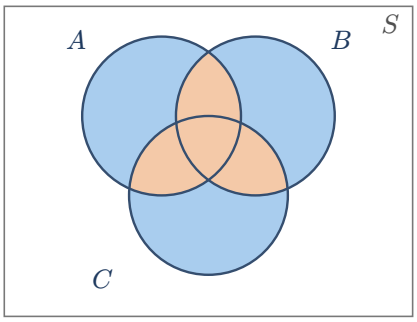
$$P(B \cap A^c) = P(B) - P(A \cap B),$$
$$P(B^c \cap A) = P(A) - P(A \cap B).$$



Proof. B is the disjoint union of $B \cap A$ (amber) and $B \cap A^c$ (blue). By Axiom 3, $P(B) = P(A \cap B) + P(B \cap A^c)$; rearrange. The second line is the first with the roles of A and B exchanged. □

Theorem 5. For three events A, B and C,

$$\begin{aligned} P(A \cup B \cup C) = {} & P(A) + P(B) + P(C) \\ & - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ & + P(A \cap B \cap C). \end{aligned}$$

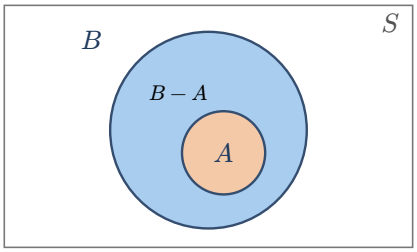


Proof. Apply Theorem 4 to the two events $A \cup B$ and C, then again to $A \cup B$, and expand $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$ with Theorem 4 once more. Collecting terms gives the formula. □

The pattern is **add the singles, subtract the pairs, add back the triple**. In the figure, the amber regions are the ones $P(A) + P(B) + P(C)$ counts more than once. A two-set overlap is counted twice by the singles and removed once by its pair. The centre is counted three times by the singles, removed three times by the pairs, and restored once by the triple. Either way, every region ends up counted exactly once, as it should.

Theorem 6. If $A \subset B$, then

(i) $P(A) \leq P(B)$;
(ii) $P(B - A) = P(B) - P(A)$.



Proof. Since $A \subset B$, B is the disjoint union of A and $B - A$, so $P(B) = P(A) + P(B - A)$. That is (ii). Since $P(B - A) \geq 0$ by Axiom 1, (i) follows. □

Watch out. $P(B - A) = P(B) - P(A)$ holds **only** when $A \subset B$. In general the right formula is Corollary 1, $P(B - A) = P(B) - P(A \cap B)$ — and when $A \subset B$ the two agree, because then $A \cap B = A$.

5.1 Worked example

Example 4. If $P(A) = 0.3$, $P(B) = 0.2$ and $P(A \cap B) = 0.1$, determine the following probabilities:

(a) $P(A^c)$ (b) $P(A \cup B)$ (c) $P(A^c \cap B)$
(d) $P(A \cap B^c)$ (e) $P((A \cup B)^c)$ (f) $P(A^c \cup B)$

Solution.

(a) Theorem 3:

$$P(A^c) = 1 - P(A) = 1 - 0.3 = 0.7.$$

(b) Theorem 4:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.3 + 0.2 - 0.1 = 0.4.$$

(c) Corollary 1:

$$P(A^c \cap B) = P(B) - P(A \cap B) = 0.2 - 0.1 = 0.1.$$

(d) Corollary 1:

$$P(A \cap B^c) = P(A) - P(A \cap B) = 0.3 - 0.1 = 0.2.$$

(e) Theorem 3 applied to the event $A \cup B$, using (b):

$$P((A \cup B)^c) = 1 - P(A \cup B) = 1 - 0.4 = 0.6.$$

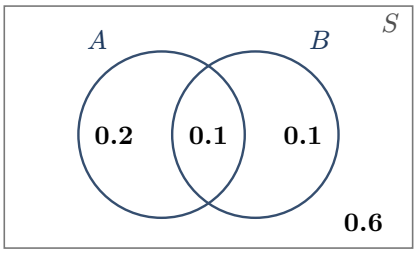
(f) Theorem 4 applied to A^c and B, using (a) and (c):

$$P(A^c \cup B) = P(A^c) + P(B) - P(A^c \cap B) = 0.7 + 0.2 - 0.1 = 0.8.$$

5.1.1 The region method

Every answer above can be read off one picture. A two-event Venn diagram has four regions; find the probability of each, write it in, and every question becomes "add up the regions it covers".

- **both**: $P(A \cap B) = 0.1$, given;
- **only A**: $0.3 - 0.1 = 0.2$;
- **only B**: $0.2 - 0.1 = 0.1$;
- **neither**: $1 - (0.2 + 0.1 + 0.1) = 0.6$.



Now (f), $A^c \cup B$, is everything except "only A": $1 - 0.2 = 0.8$, which agrees with the formula. The four region probabilities must sum to 1 — a free check.

> **Note.** The original writes part (e) as $P(A \cup B)^c$. Read literally that is the complement of a number, which means nothing; what is meant is $P((A \cup B)^c)$, the probability that **neither** event occurs. Write the brackets in.

Exercise set 2

2.1 If $P(A) = 0.5$, $P(B) = 0.4$ and $P(A \cap B) = 0.2$, find $P(A \cup B)$, $P(A^c \cap B^c)$, $P(A \cap B^c)$, and the probability that exactly one of A, B occurs.

2.2 If $A \subset B$, $P(A) = 0.25$ and $P(B) = 0.6$, find $P(B - A)$, $P(A \cap B)$ and $P(A \cup B)$.

2.3 Three events have $P(A) = P(B) = P(C) = 0.4$, each pairwise intersection has probability 0.1, and $P(A \cap B \cap C) = 0.05$. Find $P(A \cup B \cup C)$, and the probability that none of them occurs.

2.4 $P(A) = 0.3$ and $P(B) = 0.8$. Show that $P(A \cap B) \geq 0.1$. (Hint: $P(A \cup B) \leq 1$.)

2.5 Prove Corollary 1 from the axioms without looking at the proof above.

6 Mutually exclusive events

Mutually exclusive means **cannot happen at the same time**. For example:

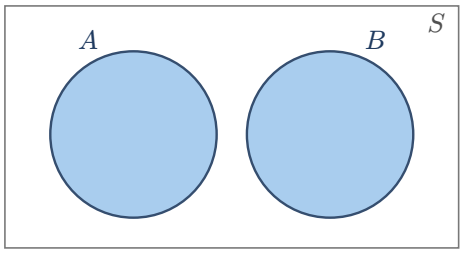
- turning left and turning right are mutually exclusive — you cannot do both at once;
- tossing a coin, heads and tails are mutually exclusive.

Definition 3.7 — Mutually exclusive. Two events A and B are **mutually exclusive** (or **disjoint**) if

$$A \cap B = \Phi.$$

For mutually exclusive events,

$$P(A \cap B) = 0 \quad \text{and} \quad P(A \cup B) = P(A) + P(B).$$



Note. The original states the definition through the two probability facts. They are **consequences** of $A \cap B = \Phi$ — Theorem 2 gives the first, Axiom 3 the second — and the definition itself is about the sets: no outcome belongs to both. Equivalently, the addition rule (Theorem 4) loses its subtraction term, because there is no overlap to double-count.

Example 5. What is the probability of getting a total of 8 or 10 when a pair of fair dice is tossed?
Solution. Let A be the event that a total of 8 occurs and B the event that a total of 10 comes up. The sample space has $N = 6 \times 6 = 36$ equally likely outcomes; in the grid, A is shaded green and B amber:

1st \ 2nd	1	2	3	4	5	6
1	(1, 1)	(1, 2)	(1, 3)	(1, 4)	(1, 5)	(1, 6)
2	(2, 1)	(2, 2)	(2, 3)	(2, 4)	(2, 5)	**(2, 6)**
3	(3, 1)	(3, 2)	(3, 3)	(3, 4)	**(3, 5)**	(3, 6)
4	(4, 1)	(4, 2)	(4, 3)	**(4, 4)**	(4, 5)	**(4, 6)**
5	(5, 1)	(5, 2)	**(5, 3)**	(5, 4)	**(5, 5)**	(5, 6)
6	(6, 1)	**(6, 2)**	(6, 3)	**(6, 4)**	(6, 5)	(6, 6)

$$A = \{(2,6), (3,5), (4,4), (5,3), (6,2)\}, \quad B = \{(4,6), (5,5), (6,4)\},$$

so

$$P(A) = \frac{5}{36} \quad \text{and} \quad P(B) = \frac{3}{36}.$$

The probability of getting a total of 8 or 10 is $P(A \cup B)$. The events A and B are mutually exclusive, since a total of 8 and a total of 10 cannot both occur on the same toss — no cell is both colours. Therefore

$$P(A \cup B) = P(A) + P(B) = \frac{5}{36} + \frac{3}{36} = \frac{8}{36} = \frac{2}{9}.$$

Watch out. Check for overlap **before** adding. "A total of 8 or a double" is not mutually exclusive — $(4,4)$ is both — and simply adding $5/36 + 6/36$ counts that cell twice. The right answer is $(5 + 6 - 1)/36 = 10/36$, Theorem 4 in action.

7 Working with the rules

The remaining examples are all the same kind of problem: some probabilities are given, others are asked for, and the theorems of §5 are the equations connecting them. Write down which theorem links what you have to what you want, then solve.

Example 6. If $P(A) = 1/4$, $P(B^c) = 2/3$ and $P(A \cap B^c) = 1/6$, find $P(A \cup B^c)$ and $P(A \cup B)$.
Solution. First recover $P(B)$ and $P(A \cap B)$, which every formula below needs.

By Theorem 3,

$$P(B^c) = 1 - P(B) \Rightarrow 1 - P(B) = \frac{2}{3} \Rightarrow P(B) = \frac{1}{3}.$$

By Corollary 1,

$$P(A \cap B^c) = P(A) - P(A \cap B) \Rightarrow \frac{1}{4} - P(A \cap B) = \frac{1}{6} \Rightarrow P(A \cap B) = \frac{1}{12}.$$

Now Theorem 4, twice:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{1}{4} + \frac{1}{3} - \frac{1}{12} = \frac{1}{2},$$

$$P(A \cup B^c) = P(A) + P(B^c) - P(A \cap B^c) = \frac{1}{4} + \frac{2}{3} - \frac{1}{6} = \frac{3}{4}.$$

Example 7. If $P(A \cup B) = 2/3$, $P(A^c \cap B) = 1/6$ and $P(A \cap B^c) = 1/3$, find $P(A)$ and $P(B)$.
Solution. Write each given probability as an equation in the unknowns $P(A)$, $P(B)$, $P(A \cap B)$:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{2}{3} \quad (1)$$

$$P(A^c \cap B) = P(B) - P(A \cap B) = \frac{1}{6} \quad (2)$$

$$P(A \cap B^c) = P(A) - P(A \cap B) = \frac{1}{3} \quad (3)$$

Subtracting (2) from (1) leaves $P(A)$ alone:

$$P(A) = \frac{2}{3} - \frac{1}{6} = \frac{1}{2}.$$

Substitute in (3): $1/2 - P(A \cap B) = 1/3$, so $P(A \cap B) = 1/6$. Substitute in (2): $P(B) - 1/6 = 1/6$, so

$$P(B) = \frac{1}{3}.$$

> **Note.** **By regions.** $A \cup B$ is the three regions only A, both, only B. Two are given, so the third is
>
> $$P(A \cap B) = \frac{2}{3} - \frac{1}{3} - \frac{1}{6} = \frac{1}{6},$$
>
> and then $P(A) = 1/3 + 1/6 = 1/2$ and $P(B) = 1/6 + 1/6 = 1/3$. One line instead of three equations — and a useful check on the algebra.

Example 8. A student is taking two courses, computer science and math. The probability that the student will pass the computer science course is 0.65, the probability that they will pass the math course is 0.70, and the probability that they will pass both courses is 0.45. What is the probability of passing **at least one**?

Solution. Let A be the event that the student passes computer science and B the event that they pass math. Then

$$P(A) = 0.65, \quad P(B) = 0.70, \quad P(A \cap B) = 0.45.$$

"At least one" is the union, so by Theorem 4,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.65 + 0.70 - 0.45 = 0.90.$$

Note. **Going further** — the same three numbers answer the other standard questions. Passing **neither**: $1 - 0.90 = 0.10$. Passing **exactly one**: $0.90 - 0.45 = 0.45$. Passing **math only**: $0.70 - 0.45 = 0.25$. Notice that $P(A) + P(B) = 1.35 > 1$: the events cannot be mutually exclusive, and the overlap is what brings the total back below 1.

Example 9. If A and B are two mutually exclusive events with $P(A) = 0.4$, $P(B) = K$ and $P(A \cup B) = 0.6$, find K.

Solution. Since A and B are mutually exclusive, $P(A \cap B) = 0$ and

$$P(A \cup B) = P(A) + P(B) \Rightarrow 0.6 = 0.4 + K \Rightarrow K = 0.2.$$

7.1 The method, in three lines

Step	What you write	Where it goes wrong
1. Translate	Each phrase as a set: "at least one" $= A \cup B$, "neither" $= (A \cup B)^c$, …	Reading "or" as exclusive; dropping the brackets in $P((A \cup B)^c)$
2. Pick the rule	The theorem connecting the givens to the target — or fill in the four regions	Adding probabilities of events that overlap
3. Compute, then check	The number, and $0 \le P \le 1$; regions sum to 1	A "probability" of 1.35 or -0.05, left unnoticed

Exercise set 3

3.1 Two fair dice are tossed. Find the probability that the total is 7 or 11. Are the two events mutually exclusive?

3.2 A card is drawn from a standard deck. Find the probability that it is a heart or a face card (Jack, Queen, King).

3.3 In a class, 60% of students study English, 45% study French and 25% study both. A student is chosen at random. Find the probability that they study neither language, and that they study exactly one.

3.4 If $P(A \cup B) = 0.8$, $P(A^c \cap B) = 0.3$ and $P(A \cap B^c) = 0.4$, find $P(A \cap B)$, $P(A)$ and $P(B)$.

3.5 A and B are mutually exclusive with $P(A) = 0.35$ and $P(B) = 0.45$. Find $P(A^c)$, $P(A \cap B)$, $P(A \cup B)$ and $P(A^c \cap B^c)$.

3.6 A die is tossed once. Let A be "an even number" and B be "a prime number". Find $P(A \cup B)$ twice — once with Theorem 4, once by listing $A \cup B$ — and check that the answers agree.

3.7 Can two mutually exclusive events have $P(A) = 0.6$ and $P(B) = 0.7$? Explain.